



Society of Petroleum Engineers

SPE-228884-MS

Cost-Effective Bottomhole Flow Diagnostics in Mature Assets Using Dissolvable Ball with Nanoparticle Tracers: A Case Study from East Texas

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/228884-MS](https://doi.org/10.2118/228884-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Managing mature oil and gas wells under a Proved Developed Producing (PDP) investment model requires cost-effective diagnostics to guide intervention strategies and maximize production. In the Tennessee Colony Field located in Anderson County, East Texas, Petralis Energy Resources LLC deployed an innovative production flow diagnostic technology using nanoparticle tracers encapsulated within dissolvable alloy spheres or balls to evaluate bottomhole flow in the Flockhart #1 mature vertical well. The dissolvable alloy balls, available in multiple sizes (2.0-in., 2.5-in., 3.5-in., or 4.5-in. outer diameter), are engineered to withstand extreme downhole conditions, including pressures up to 7,500 psi and temperatures up to 500°F, while the tracers themselves remain stable at temperatures up to 2,000°F. Deployment involves dropping or pumping the balls to the bottom of vertical wells or the toe of horizontal wells, where they dissolve within approximately 72 hours, releasing non-radioactive, non-hazardous nanoparticle tracers into the production stream. These tracers are detected with ultra-high sensitivity sub-atomic analytical methods, providing near real-time flow diagnostics without the need for costly downhole modifications or interventions. The precision-machined, highly engineered dissolvable alloy shell of the ball functions as a housing for tracer deployment. This sphere is easily dropped or pumped to the bottom of a vertical well or the toe of an extended-reach horizontal section for deployment. The case study of the Flockhart #1 well demonstrates the technology's ability to deliver actionable, high-resolution flow data with minimal disruption and at a fraction of the cost of traditional methods. Tracer data recovery was robust, well above the target detection threshold and confirmed restored production flow from targeted intervals following acid stimulation, validating the workover and aligning with Petralis' cost-sensitive strategy. Compared to conventional techniques, this novel approach minimizes operational complexity and expense while creating new opportunities for downhole production monitoring, and enhanced well management in mature fields, representing a breakthrough advancement in flow diagnostics for oil and gas producers.

Introduction

Mature oil and gas wells account for about 7 percent of U.S. hydrocarbon production and represent approximately 70 percent of total producing wells in the U.S ([U.S. EPA, 2025](https://www.epa.gov/energy/energy-production-and-consumption)). These low-rate producers,

often categorized as "stripper wells," remain economically viable under a Proved Developed Producing (PDP) investment model that emphasizes capital efficiency and predictable cash flow. Despite their declining output, these wells offer significant opportunities for incremental recovery when managed with financial discipline and careful workover technology selection and execution. However, field operators face persistent challenges in assessing well performance after workovers, re-completion, or re-stimulation. Conventional diagnostics—such as production logging tools (PLT) or fiber optic systems—often prove cost-prohibitive and intrusive, especially in mature fields where operating margins are narrow. This underscores the need for low-cost and risk-free production flow diagnostic methods that align with the economic realities of legacy well operations.

This paper outlines a field-proven ultrahigh-resolution nanoparticle tracer technology tailored for mature field applications. The method offers non-intrusive deployment, high sensitivity, and cost-effective scalability, enabling asset teams to quantify flow contribution, validate interventions, and optimize reservoir management across PDP portfolios. A case study from Tennessee Colony Field in Anderson County, East Texas, illustrates how this technology supports practical, value-oriented decision-making in the context of mature well revitalization.

The PDP Model—A Cost-Conscious Strategy to Maximize Mature Well Investment Return

The PDP model is a strategic framework for maximizing returns from mature oil and gas assets by acquiring wells with established production histories and low decline rates in the long-tail production stage in fields like Tennessee Colony. This approach generates cash flow to fund operations and targeted interventions. The model prioritizes cost-effective workovers, such as acid stimulation, to address near-wellbore damage, enhance permeability, and enhance production, extending the economic life of mature wells. [Jain \(2024\)](#) refers to mature fields as "reservoirs of opportunity", reflecting an optimistic perspective on the economic and technological potential of aging oil and gas fields. A balanced view would necessarily acknowledge the challenging tight economics and environmental concerns that need to be managed.

The PDP model's success in East Texas is supported by the region's low decline rates, which allow wells to maintain stable production with minimal investment. Workovers enable operators to target specific production issues, such as near-wellbore damage or water cut, ensuring resources are allocated efficiently to maximize cash flow and estimated ultimate recovery (EUR).

In East Texas, workovers are critical for revitalizing mature wells in carbonate reservoirs like the Rodessa and Pettet. Acid stimulation dissolves obstructions (e.g., drilling mud, scale, organic deposits) and creates flow channels, improving connectivity between the reservoir and wellbore. The economic viability of the PDP model depends on minimizing capital expenditure (CAPEX) and operational expenditure (OPEX) while maximizing estimated ultimate recovery (EUR).

Traditional diagnostics cost such as production logging tools (\$150,000 to \$300,000 per well) and fiber optic DTS/DAS systems (\$250,000 to \$1,500,000 per well) are known to be costly and typically out of reach by operators of mature wells. Other techniques, including chemical and radioactive tracers require special handling accompanied by regulatory requirements that add cost and complexity and similarly of reach for highly cost-sensitive PDP-focused operators. Cost-effective production diagnostic technology, like nanoparticle tracer technology described here, aligns with the PDP model's emphasis on operational efficiency, simplicity, and economic discipline. Nanoparticle tracer technology enables operators to validate interventions and optimize production without significant financial outlay.

Petalis Energy Partner's PDP Assets and Investment Model

Petalis Energy Resources is a Houston-based private upstream oil & gas company established in 2019. The company focuses on acquiring, operating, and optimizing underperforming conventional oil and gas

assets across East Texas including the Tennessee Colony, Gladewater, Nacogdoches Creek, and Chapel Hill regions.

Petralis acquires lightly capitalized, mature conventional oil and gas assets and applies low-risk interventions (workovers and stimulation) to enhance production and extend the economic life of the mature wells in its portfolio. Their strategy emphasizes tax- advantaged, low-decline asset optimization for investors seeking predictable cash flow from low-operational-cost legacy wells.

Petralis owns a modest East Texas asset base of mature wells producing approximately 105 Kbbbls of oil and over 5 MMCF of gas since 2019, with daily operations delivering around 100 BOPD and 4.5 MMCFD gas. The company emphasizes repeatable and low decline PDP models, technical risk reduction, and investor-aligned cash flow through low-cost intervention strategies and production management.

The Flockhart #1 well referred to in this case study is in the Tennessee Colony Field in Anderson County in East Texas, approximately 14 miles northwest of the city of Palestine, East Texas.

Historical and Geologic Context of Oil and Gas Production in Anderson County, Texas

Anderson County, situated in the East Texas Basin, was a cornerstone of hydrocarbon production since the early 20th century. The discovery of the enormous East Texas Oil Field in 1930, producing over 5.42 billion barrels of oil, established the region as a key oil and gas development center ([Texas Historical Commission, 2022](#)). The Tennessee Colony Field targets carbonate reservoirs in the Rodessa, Pettet, and James Limestone formations as noted by [Ely \(1950\)](#). Deeper wells, often exceeding 9,000 ft, target these carbonate reservoirs which sit below the much shallower Woodbine Formation that was the primary formation that drove production in the early exploitation of the East Texas Oil Field.

The Rodessa Formation, a Cretaceous limestone, produces oil from its First Rodessa zone and gas with condensate from the Second Rodessa zone. The Pettet Formation, also a limestone, is a significant oil producer, while the James Limestone dominates in the nearby Fairway Oil Field, with exploration extending into Anderson County. These formations, formed in a shallow marine environment approximately 100 million years ago, exhibit complex facies changes due to depositional variations and diagenesis, resulting in heterogeneous porosity and permeability.

Oil and gas production in Anderson County peaked in the mid-20th century, driven by conventional vertical well technology. By the late 20th and early 21st centuries, many wells entered a mature phase, characterized by declining production rates due to reservoir pressure depletion, near-wellbore damage, and increased water production. Operators responded with workover programs, including acid stimulation and hydraulic fracturing, to remove near-wellbore damage and restore flow. The introduction of advanced flow diagnostics, such as chemical liquid tracers and fiber optic systems, improved reservoir management historically but often at high costs, contributing to the drive to develop far more cost-effective alternatives highlighted by the nanoparticle tracer deployment discussed here.

Top oil and gas producers in Anderson County include many smaller, independent companies. These types of operators often specialize in acquiring and managing mature assets, including stripper wells, where the focus is on cost-efficiency to maintain marginal production. Their business model is typically centered on maximizing recovery from existing wells rather than large-scale, high-volume, and costly drilling programs.

Operating stripper wells requires tight budget management. The lower production volumes mean that profit margins per barrel or Mcf are smaller, necessitating highly efficient operations to remain economically viable. Operators often employ specialized techniques to extend the life of these wells, including acid stimulation jobs and clean-outs like what was conducted by Petralis in the Flockhart #1 well. Due to the low operating costs, stripper well operators in Anderson County, TX, and across Texas, can often be profitable despite relatively low oil and gas production.

These cost-conscious operators are now benefiting from simple, low-cost technology that is emerging thanks to the properties and characteristics of nanoparticles and their novel applications in oil and gas production flow diagnostics.

Nanoparticle Properties Unique Characteristics

A nanoparticle is a particle of matter with at least one dimension between 1 and 100 nanometers (nm). To put this perspective, the typical 100 Mesh Frac Sand (proppant) is about 149,000 nanometers in diameter, and a single atom is 0.1 to 0.5 nm in size. Due to their special properties, nanoparticles are finding broad application in the oil and gas industry, as noted by [Ko and Huh \(2019\)](#).

Table 1—Nanoparticles in Perspective

Item	(nm)	Size (pm)	Description
Atom	0.1	0.0001	Approximate diameter of a single atom, the Bohr radius, and distance from the proton to the electron
Molecule	0.1-10	0.0001-0.01	Typical range of small organic or inorganic molecules
Nanoparticles	1-100	0.001-0.1	Scale of particles deployed in spheres used for production flow diagnostics
Typical 100-mesh Frac Sand (Proppant)	149,000	149	Size classification of a common frac sand (100 mesh)

nm=nanometer, Lim=micrometer

As a result of their size, nanoparticles exhibit properties different from both their individual atom and bulk material counterparts, as noted by ([Cao and Wang, 2011](#)). As materials are engineered at the nanoscale, their surface area-to-volume ratio increases significantly, and they begin to exhibit quantum confinement effects, which alter their optical, electronic, and magnetic properties in ways that differ from their bulk counterparts ([Baig et al., 2021](#)). These phenomena can be exploited to design nanoparticles with highly specific and detectable behaviors. Engineered nanoparticle tracers can be synthesized with controlled size, shape, and composition to create distinct elemental or isotopic signatures that remain detectable at extremely low concentrations, often in the parts-per-billion (ppb) range.

Detection is achieved through high-sensitivity sub-atomic analytical techniques that rely on elemental or isotopic fingerprinting, where each tracer is identified by its unique atomic mass or spectral signature. In contrast to conventional tracers, these nanoparticles can be formulated for extended stability, low chemical reactivity, minimal adsorption onto reservoir rock surfaces, and ultrahigh-temperature application. This allows for longer residence time in the formation and clearer signal detection in complex multiphase production fluids. The result is a tracer system that offers a definitively detectable, robust, and unique fingerprint that is also well-suited for long-duration reservoir surveillance and flow diagnostics.

How Nanoparticles Compare to Other Tracer Types

A variety of tracers are used in oil and gas wells to monitor production flow, each offering distinct benefits and limitations depending on the application environment and diagnostic goals. Tracer longevity, signal clarity and resolution, handling requirements, and environmental stability are key differentiators when selecting a tracer system. The table below compares many of the most used tracer types, including nanoparticles, conventional liquid chemicals, perfluorocarbons (PFCs), rare-earth elements (REEs), gases, and radioisotopes, highlighting their typical operational durations and primary characteristics. The specific tracer type selected carries with it a defined set of unique features, performance capabilities, and often, challenges that operators carefully evaluate. Nanoparticle tracers exhibit many characteristics that are highly

desirable when deploying tracers for production flow diagnostics and offer operators a new, low risk technology to consider and evaluate.

Table 2—Comparing Tracers Used in Oil and Gas Wells for Production Flow Diagnostics

Tracer Type	Typical Duration	Characteristics, Strengths, Limitations
Nanoparticle tracers (used in this case study)	Up to ~3 months	Inert, non-radioactive, and engineered for thermal and chemical stability under harsh conditions. Designed for long-term deployment, controlled release, low adsorption, and unique tagging for unambiguous detection.
Conventional liquid chemical tracers (e.g., fluorinated benzoates, alcohols, iodides)	Days to weeks	Susceptible to dilution, dispersion, and adsorption onto formation rock, limiting persistence. Commonly used in short-duration flowback or interwell tests. May require special handling and regulatory requirements depending on formulation.
Perfluorocarbon (PFC) tracers	Hours to days	Chemically inert but highly volatile. Require careful containment and handling. Prone to leakage and environmental concerns.
Rare-earth element (REE) tracers	Weeks to ~1 month (longer when encapsulated)	Moderately stable due to low reactivity. Can be chemically susceptible and adsorbed by rock. Background signal interference possible due to natural REE presence in some formations.
Gas tracers (e.g., SFe, noble gases)	Hours to weeks	Prone to loss via phase separation, diffusion, or production cycling. Limited duration and higher mobility can reduce diagnostic resolution.
Radioactive tracers (commonly used isotopes)	Days to up to ~3 months (depends the half-life of the	Offer strong signal detectability but involve strict regulatory requirements, special logistics, and environmental risk considerations.

Comparison with Conventional Diagnostics

Conventional diagnostic methods often present significant limitations in mature fields, particularly concerning their high cost, operational complexity, and data resolution. For instance, production logging tools (PLT), including slim PLT, are widely used to obtain direct quantitative measurements of flow rates and fluid types at various depths. While valuable for providing a snapshot of well performance, their deployment necessitates wireline or slickline intervention, shut-in production, making them an intrusive solution that interrupts production and incurs substantial costs, typically ranging from \$150,000 to \$300,000 per well for each survey.

Similarly, fiber optic Distributed Temperature Sensing (DTS) and Distributed Acoustic Sensing (DAS) systems (which encompass passive acoustic sensing applications) offer the advantage of continuous, real-time data acquisition along the wellbore. However, the upfront capital expenditure for these systems is considerable, known to range from \$250,000 to \$1,500,000 per well for permanent installations or even temporary deployments utilizing methods like fiber-in-capillary for protection. Furthermore, DAS provides indirect measurements of flow through acoustic signatures, requiring complex interpretation to translate into quantitative flow rates from specific intervals. The intrinsic high cost and intricate installation procedures associated with these advanced fiber optic systems often render them economically impractical for cost-sensitive PDP-focused operators managing mature assets.

Other established techniques, such as chemical liquid tracers, also present notable drawbacks. While offering strong signal detectability and being relatively well-understood, they typically require multiple formulations for different fluid types, increasing logistical complexity and overall project costs (often around \$150,000 per well). Moreover, most chemical liquid tracers are susceptible to rapid dilution, dispersion, and adsorption onto reservoir rock surfaces, which severely limits their monitoring duration

to days or weeks. This rapid dissipation, coupled with their propensity for false positives, particularly in complex fracture networks where tracers may penetrate microscopic fissures or micro fractures rather than true flow paths, often reduces their diagnostic resolution and reliability for long-term production assessment.

Radioactive tracers, while offering strong signals, impose strict regulatory requirements, special handling logistics, and significant environmental risk considerations, making them less desirable. Finally, microseismic surveys are primarily designed to measure fracture geometry and propagation, rather than directly assessing production flow, thus limiting their utility for production diagnostics in this context. [Tayyib et al. \(2019\)](#) reviewed the use of chemical liquid tracers, alongside other methods like radioactive tracers.

In stark contrast, nanoparticle tracer technology effectively overcomes these limitations by providing stable, long-term monitoring (up to three months with multiple deployments). Its ability to provide direct evidence of flow from targeted intervals, without the need for invasive interventions or the high costs associated with conventional methods, represents a significant leap forward. This novel approach minimizes operational complexity and expense while delivering high-resolution, unambiguous flow diagnostics, making it an ideal, cost-effective solution for optimizing production and enhancing well management in mature fields under the PDP model.

Nanoparticle Tracer Deployment Methodology

The bottomhole production diagnostic technology employs nanoparticle tracers encapsulated in a dissolvable alloy hollow ball, designed to assess production flow from the bottom of vertical wells or the toe of horizontal wells. It is applicable to newly completed, mature onshore, or offshore wells, supporting oil, gas, water, or geothermal production. The alloy ball, available in 2.0-in., 2.5-in., 3.5-in., or 4.5-in. outer diameters, is precision-machined to withstand differential pumping pressures up to 7,500 psi and temperatures up to 500°F, with tracers stable at temperatures up to 2,000°F. The non-radioactive, non-hazardous tracers are engineered for ultra-high sensitivity, ensuring precise flow quantification.

Deployment is non-intrusive and risk-free, requiring no subsurface or surface modifications. In vertical wells, the ball is simply dropped into the well from the surface and relies on gravity to reach the bottom. In horizontal wells, the sphere is pumped to the toe using standard pump-down or frac equipment. The ball dissolves within approximately 72 hours upon contact with freshwater, brine, frac fluid, or production fluids, releasing the tracers into the flow stream. Samples are collected from the flowline and analyzed using proprietary ultra-high resolution sub-atomic detection methodologies, providing near real-time results. The technology's flexibility allows for single or multiple ball deployments, enabling extended monitoring periods of up to three months, unlike chemical liquid tracers that dissipate more quickly in comparison.

The analytical lab and nanoparticle tracer analysis process reduces turnaround time from 2 weeks to 1-2 days. The tracers' stability ensures continuous monitoring, providing insights into flow behavior over time. This contrasts with traditional methods, which often require invasive interventions or provide only short-term snapshots of production.

Flockhart #1 Well Workover and Tracer Deployment

The Flockhart #1 well, a mature vertical well in the Tennessee Colony Field, produces from perforations at 8,947-8,952 ft, 8,957-8,979ft, and 8,997-9,010 ft. In February 2024, the well underwent a workover to address near-wellbore damage and restore production. The procedure included:

1. Removal of Production Equipment: The production tubing (2-7/8" 6.5# N-80 EUE 8rd) and sucker rods were pulled out, and the ABRA control dynamometer was removed for return to Houston.
2. Well Cleaning: A 4-3/4" rock bit was run to tag the bottom at 9,500 ft (plug-back total depth). KCL water was circulated to clean the perforations and near-wellbore zone in the Pettet formation.

3. Acid Stimulation: A packer was set at 9,050 ft, and 25 barrels of 15% NEFE acid were pumped to dissolve scale, drilling mud, and precipitates. The packer was released, and tools were pulled out.
4. Tracer Deployment: Two dissolvable alloy balls containing ultrahigh-resolution nanoparticle tracers were dropped to the bottom, targeting the deepest perforated intervals.
5. Recompletion: The well was recompleted with 2-7/8" tubing, a tubing anchor set with 12,000 lbs. tension, and a bull plug at 9,278.31 ft. Production resumed, and flowline sampling began in March 2024.

Samples were analyzed to measure tracer concentrations in parts per billion (ppb) in water and oil phases, with a target range of 0.2-0.3 ppb indicating successful flow. The deployment was designed to confirm whether the acid stimulation restored flow from the targeted intervals, providing direct evidence of the workover's effectiveness.

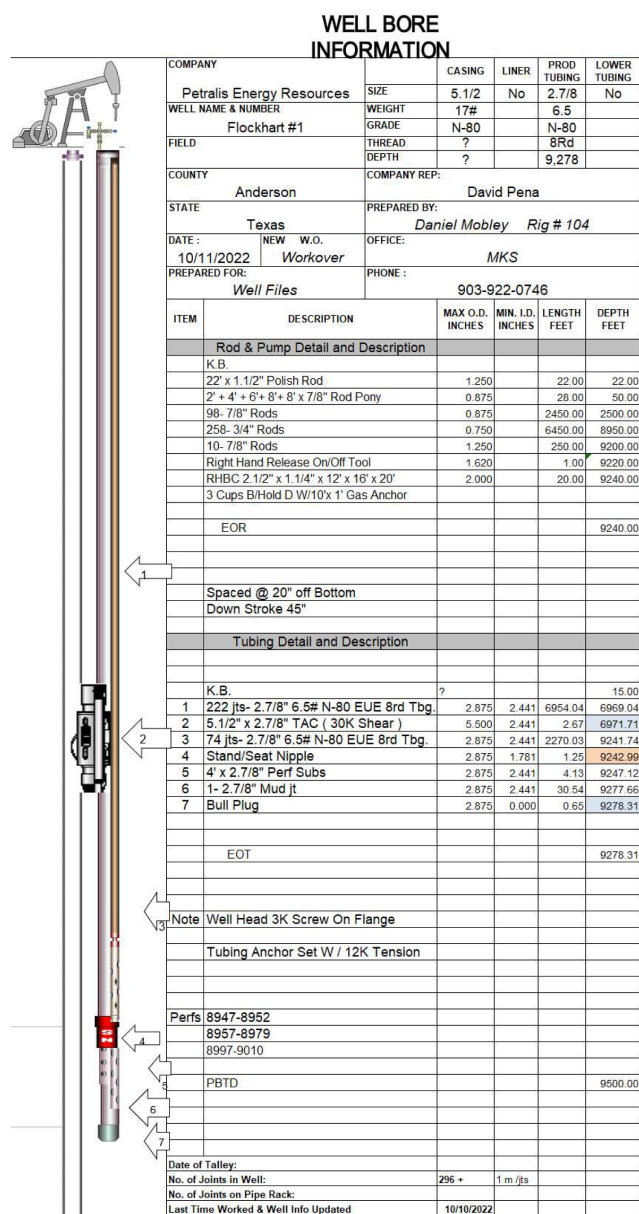


Figure 1—Wellbore Schematic, Flockhart #1, Anderson County, TX

Results

The deployment in the Flockhart #1 well confirmed restored production from the Pettet formation's perforated intervals. Tracer concentrations of 2.7 ppb in water and 13.1 ppb in oil were measured, exceeding the target range, indicating robust multiphase flow from the bottom of the well. Oil production increased from 4.7 to 7.9 BOPD or 68%, when comparing the six months before the workover to the six months after.

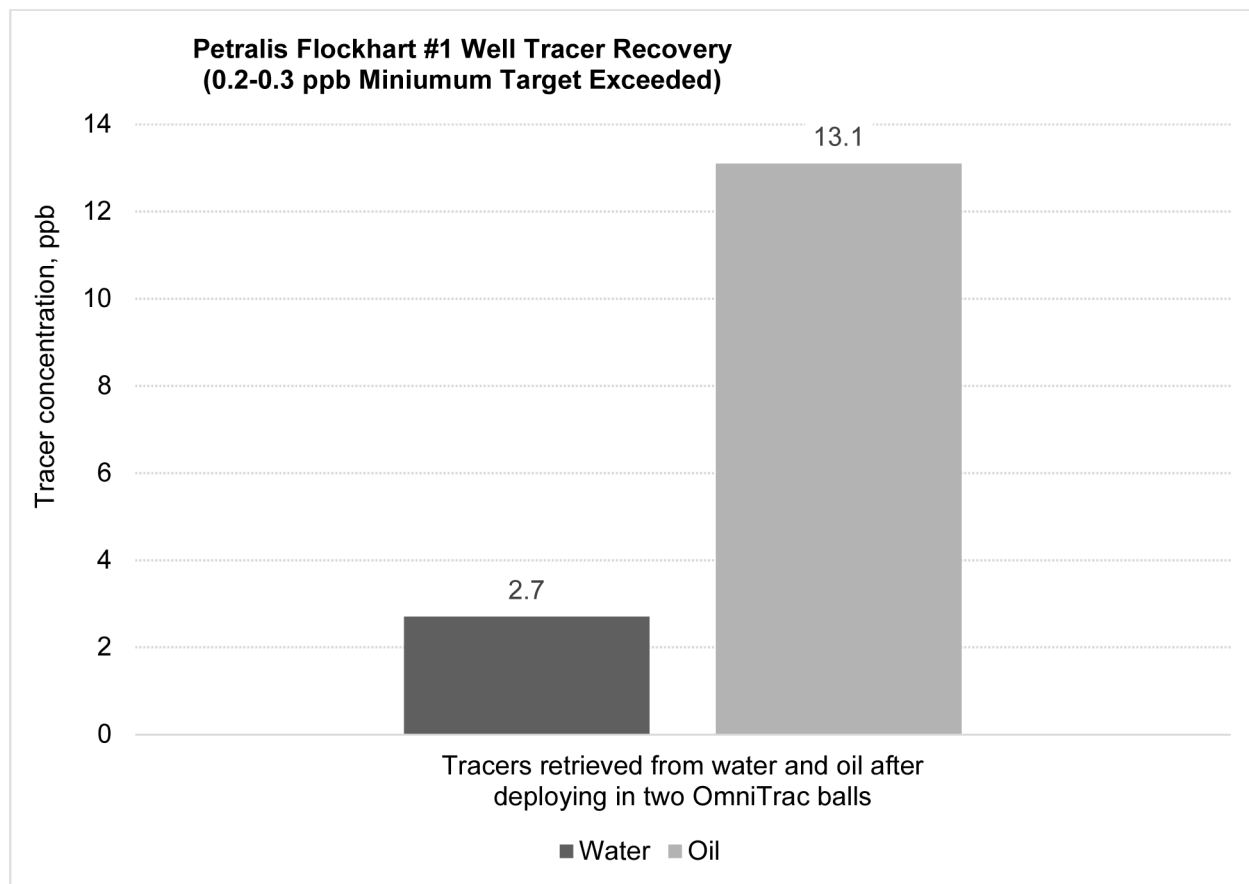


Figure 2—Tracer Recovery and Detection Target in Anderson County, East Texas Flockhart #1 Well.

These results validated the acid stimulation's success in clearing near-wellbore damage and enhancing flow permeability, as the tracers were carried to the surface with production fluids. The presence of tracers in both water and oil phases confirmed flow from the targeted intervals, ruling out blockages or isolation. The absence of tracers would have indicated a need for further intervention, such as additional cleaning or re-perforation. The nanoparticle tracer technology provided high-resolution data with minimal disruption, at a cost significantly lower than production logging tools or fiber optic systems, aligning with the PDP model's cost sensitivity and the desire for simplicity.

The results also provided insights into the Pettet formation's flow characteristics, confirming that the acid stimulation created or enlarged flow channels, improving connectivity between the reservoir and wellbore. This was particularly critical in a mature field where nearwellbore damage can significantly impact production.

Geological Challenges in Deeper Anderson County Wells

Deeper wells in Anderson County, often exceeding 9,000 ft, target carbonate reservoirs like the Rodessa and Pettet formations, which exhibit complex geological characteristics. Formed in a Cretaceous shallow

marine environment, these reservoirs are affected by facies changes, diagenesis, and faulting, leading to heterogeneous porosity and permeability. Deeper reservoirs encounter higher pressures (up to 7,500 psi in some cases) and structural complexities, such as natural fractures, which can enhance permeability but increase drilling and production risks. Near-wellbore damage, caused by drilling fluids, scale, or organic deposits, is prevalent in mature wells, necessitating targeted interventions like acid stimulation.

Nanoparticle tracer technology addresses these challenges by providing precise flow diagnostics from bottom of the well to confirm flow restoration. In the Flockhart #1 well, tracer data confirmed that the acid stimulation successfully mitigated near-wellbore damage, restoring flow from the Pettet formation's perforated intervals. This is particularly valuable in heterogeneous reservoirs, where flow contributions can vary significantly across short distances, requiring accurate diagnostics to optimize production.

Role of Workovers in East Texas Mature Wells

The role of data in maximizing recovery from aging reservoirs and workovers are a cornerstone of production enhancement in East Texas mature fields, particularly in carbonate reservoirs. Acid stimulation, using 15% NEFE acid, dissolves obstructions and creates flow channels, improving permeability and restoring flow. Compared to new drilling, workovers are significantly more cost-effective, with acidizing campaigns known to often cost less than \$100,000 per well.

The PDP model relies on workovers to extend the economic life of mature wells, maximizing cash flow while minimizing CAPEX and OPEX. Flow diagnostic and measurement technologies are critical to validate these interventions, ensuring resources are allocated efficiently. The nanoparticle tracer technology's affordability and precision make it an ideal complement to workovers, providing direct evidence of flow restoration without the high costs of traditional methods.

Economic and Operational Benefits of Nanoparticle Tracer Technology

Nanoparticle tracer technology offers several significant advantages for mature field management and how other more cost-conscious operators and others approach diagnostics. Foremost among these is its cost-effectiveness, as deployment costs represent merely a fraction of traditional diagnostic methods, which are known to typically range from \$150,000 for chemical liquid tracers up to \$1,500,000 for fiber optic systems. This economic advantage aligns perfectly with the Proved Developed Producing (PDP) model's emphasis on financial discipline, enabling operators to allocate resources to additional workovers or other field development activities and better inform well maintenance and production enhancement work taken. Furthermore, its non-intrusive deployment eliminates the need for any downhole or surface modifications, thereby minimizing operational risk and complexity, with the dissolvable ball enabling simple tracer deployment. The analytical sub-atomic lab also significantly contributes to operational efficiency by eliminating the need for longer offsite sample processing, reducing turnaround time from 1-2 weeks to just 1-2 days.

The novel next-generation tracer technology also provides high-resolution data, as the ultra-high sensitivity detection of the nanoparticle tracers enables reliable flow quantification in the reservoir. Its versatility is another key benefit, with the system applicable across various well types (vertical and horizontal), fluid types, and engineered for stability under harsh subsurface conditions. The nanoparticle tracer's unique properties are also facilitating additional applications, from being pre-loaded in perforation charges for perf-level diagnostics to being pre-placed in sand screens for generating production profiles, utilizing the over 220 uniquely tagged and discretely identified nanoparticle tracers available, as outlined in (Shokanov et al. 2019) and (Hendricks et al. 2025). The tracers' stability ensures reliable and high-resolution monitoring, providing insights into flow behavior over extended periods, with multiple deployments, contrasting with traditional methods that often require invasive interventions or provide much shorter-term snapshots of production.

Finally, the method offers substantial environmental benefits. The use of non-radioactive tracers eliminates the risks associated with radioactive isotopes handling and disposal, significantly reducing the overall environmental footprint of diagnostic operations. This technology also avoids many of the handling complexities and potential environmental impact concerns inherent in some traditional chemical liquid tracers, further aligning with a growing industry emphasis on ESG (Environmental, Social, and Governance) metrics. By reducing the need for invasive diagnostics and associated equipment mobilization, the technology lowers emissions and supports sustainable reservoir management. In mature fields, where economic margins are tight, such cost-effective and environmentally conscious technologies enable operators to maintain profitability while meeting evolving environmental goals. The successful adoption and favorable results of nanoparticle tracer deployment by operators in a tight PDP model investment environment indicate that more data-driven evaluations have the opportunity to become commonplace, given the technology's lower deployment hurdle, high-resolution data, cost, and relatively low execution complexity.

Broader Applications in Mature Fields

Beyond the Tennessee Colony Field, the nanoparticle tracer technology has potential applications in other mature oil and gas regions, such as the Permian Basin, Appalachian Basin, and Gulf Coast. These regions share similar challenges, including declining production and near-wellbore damage, making cost-effective diagnostics highly advantageous. The nanoparticle technology's versatility allows it to be deployed in diverse well types, from vertical mature wells to modern horizontal completions, and across various fluid types, including oil, gas, and geothermal fluids.

In carbonate reservoirs, where acid stimulation is a common intervention, the technology can confirm the effectiveness of treatments, as demonstrated in the Flockhart #1 well. In clastic reservoirs, it can assess flow contributions from complex fracture networks, providing insights into production performance. Offshore applications, where traditional diagnostics are particularly costly, could benefit from the technology's non-intrusive deployment and low operational footprint.

Economic and Operational Benefits

The economic benefits of the nanoparticle tracer technology are significant, particularly for PDP-focused operators. By reducing diagnostic costs to a fraction of traditional methods, it enables operators to allocate resources to additional workovers or field development activities. In the Flockhart #1 well, the technology's low cost and reliable, high-fidelity diagnostics compared to alternative methods allowed the operator to validate the workover without diverting funds from other operations.

Operationally, the technology simplifies reservoir management by providing rapid, actionable data. The analytical lab eliminates the need for offsite sample processing, reducing logistical complexity and turnaround time. This is particularly valuable in legacy fields like Tennessee Colony, where access to advanced flow diagnostics and measurement technologies may be limited. The ability to deploy multiple balls for extended monitoring periods further enhances operational flexibility, allowing operators to track changes in flow behavior over time and adjust strategies accordingly.

Conclusion

The deployment of nanoparticle tracers in dissolvable alloy balls in the Flockhart #1 well demonstrates a cost-effective, non-intrusive approach to bottomhole flow diagnostics in mature oil and gas fields. By confirming restored production from the Pettet formation following acid stimulation, the technology validated the workover's success and supported the operator's PDP strategy. Its high-resolution data, affordability, and versatility make it a transformative tool for managing mature assets in Anderson County,

where geological complexities and mature field challenges necessitate innovative solutions. The historical context of Anderson County's oil and gas production underscores the need for such technologies, while the PDP model highlights the importance of cost-conscious strategies in maximizing returns. Future integration with AI and cloud-based analytics is expected to further enhance the technology's impact, enabling operators to optimize production, reduce costs, and meet ESG objectives in mature fields worldwide.

Acknowledgments

The authors appreciate the effective collaboration between Petralis Energy Resources and QuantumPro, Inc technical and operation teams. Special thanks to David Pena for his leadership in the Flockhart #1 workover campaign and QuantumPro Inc. for an excellent performance onsite during deployment and sampling work.

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